Daily Algebra Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Vieta's formulas for cubics · Nested surds · Telescoping products · Lagrange/Brahmagupta identity · Factor theorem · Cyclic sums under constraints.

Section A Rapid Recognition ($\leq$15 s each)

Factorise, simplify, or evaluate instantly. No expansion.

1. Factorise completely: $x^3 - 27$.

Ans: $(x-3)(x^2+3x+9)$

2. Without expanding, evaluate $(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})$.

Ans: 2

3. Without a calculator, evaluate $999^2 - 1$.

Ans: 998 000

4. Given $a - b = 4$ and $ab = 5$, find $a^2 + b^2$.

Ans: 26

5. Expand and simplify: $(x + y)^3 - x^3 - y^3$.

Ans: $3xy(x + y)$

6. Factorise completely: $3x^2 - 75$.

Ans: $3(x - 5)(x + 5)$

7. Rationalise the denominator: $\dfrac{1}{\sqrt{3} + \sqrt{2}}$.

Ans: $\sqrt{3} - \sqrt{2}$

8. Factorise: $x^2(x-1)-(x-1)$.

Ans: $(x-1)(x-1)(x+1) = (x-1)^2(x+1)$

9. The roots of $x^2-5x+3=0$ are p and q. Find p^2+q^2.

Ans: 19

10. Simplify: $\dfrac{2^{n+3}-2^n}{2^n}$.

Ans: 7

Section B Manipulation Drills ($\leq$50 s each)

Fractions, rearranging, hidden factorisations. Elegance over brute force.

1. Simplify: $\dfrac{x}{x^2-1}+\dfrac{1}{x-1}$.

Ans: $\dfrac{2x+1}{(x-1)(x+1)}$

2. Factorise: x^4-x^2-12.

Ans: $(x^2-4)(x^2+3)=(x-2)(x+2)(x^2+3)$

3. Make r the subject of $S=\dfrac{a}{1-r}$.

Ans: $r=1-\dfrac{a}{S}=\dfrac{S-a}{S}$

4. Simplify: $\left(1+\frac{1}{x}\right)\left(1-\frac{1}{x+1}\right)$.

Ans: $1 = \frac{1+x}{x} \cdot \frac{x}{1+x}$

5. Given $x+y=3$ and $xy=1$, find x^3+y^3.

Ans: 18

6. Simplify: $\dfrac{(a+b)^2-(a-b)^2}{4ab}$.

Ans: 1

7. Factorise: $x^3 + 3x^2 - x - 3$.

Ans: $(x-1)(x+1)(x+3)$

8. Solve: $\dfrac{x-1}{x+2} = \dfrac{x+3}{x+8}$.

Ans: $x = 7$

9. Simplify: $\dfrac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}} + \dfrac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$.

Ans: $\dfrac{2(a+b)}{a-b}$

10. The cubic $x^3 - 6x^2 + 11x - 6 = 0$ has roots α, β, γ. Without solving it, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha\beta\gamma$.

Ans: $\alpha^2 + \beta^2 + \gamma^2 = 14$; $\alpha\beta\gamma = 6$

Section C Substitution & Structure ($\leq$90 s each)

Find structure before computing. Substitution is the key, not expansion.

1. Solve: $x^4 - 13x^2 + 36 = 0$.

Ans: $x = \pm 2, \pm 3$

2. Solve: $(x+1)(x+2)(x+3)(x+4) = 24$.

Ans: $x = 0$ and $x = -5$

3. Simplify: $\sqrt{6+2\sqrt{5}}$.

Ans: $\sqrt{5}+1$

4. Given $a+b+c=1$ and $a^2+b^2+c^2=1$, find $ab+bc+ca$ and hence $a^3+b^3+c^3-3abc$.

Ans: $ab+bc+ca=0$; $a^3+b^3+c^3-3abc=1$

5. Solve: $(x-1)(x-3)(x-5)(x-7)+15=0$.

Ans: $x=2,\ 6,\ 4\pm\sqrt{6}$ (all four real)

6. If $x = 2 + \sqrt{3}$, find $x^2 + \dfrac{1}{x^2}$ without expanding x^2 directly.

Ans: 14

7. Given $\dfrac{a}{b} + \dfrac{b}{a} = 3$, find $\dfrac{a^2}{b^2} + \dfrac{b^2}{a^2}$.

Ans: 7

8. If $x + y + z = 0$, simplify $\dfrac{(x^2 + y^2 + z^2)^2}{x^4 + y^4 + z^4}$.

Ans: 2

Section D Challenge (TMUA / SMC difficulty)

Each has a solution under five lines. Find the insight, not the computation.

1. Evaluate:

$$\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\frac{1}{3\cdot 4\cdot 5}+\cdots+\frac{1}{8\cdot 9\cdot 10}.$$

Ans: $\frac{11}{45}$

2. The roots of $x^2+px+q=0$ are r and s. Find, in terms of p and q, the monic quadratic whose roots are r^2+s and s^2+r.

Ans: $x^2-(p^2-2q-p)\,x+(q^2-p^3+3pq+q)=0$

3. The equation $\dfrac{x}{y+z} = \dfrac{y}{x+z} = \dfrac{z}{x+y} = k$ holds for real x, y, z with $x + y + z \neq 0$. Find all possible values of k.

Ans: Under $x + y + z \neq 0$: $k = \frac{1}{2}$ only.

4. Prove the **Brahmagupta–Fibonacci identity**:

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2.$$

Hence express 85×65 as a sum of two perfect squares in two different ways.

Ans: $85 \times 65 = 5525 = 74^2 + 7^2 = 71^2 + 22^2$.

5. Find all integers $n \geq 1$ for which $n^2 + n + 1$ divides $n^3 - 1$.

Ans: All integers $n \geq 1$.